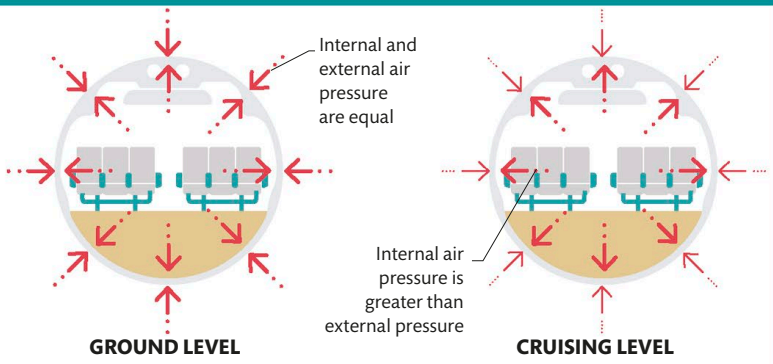
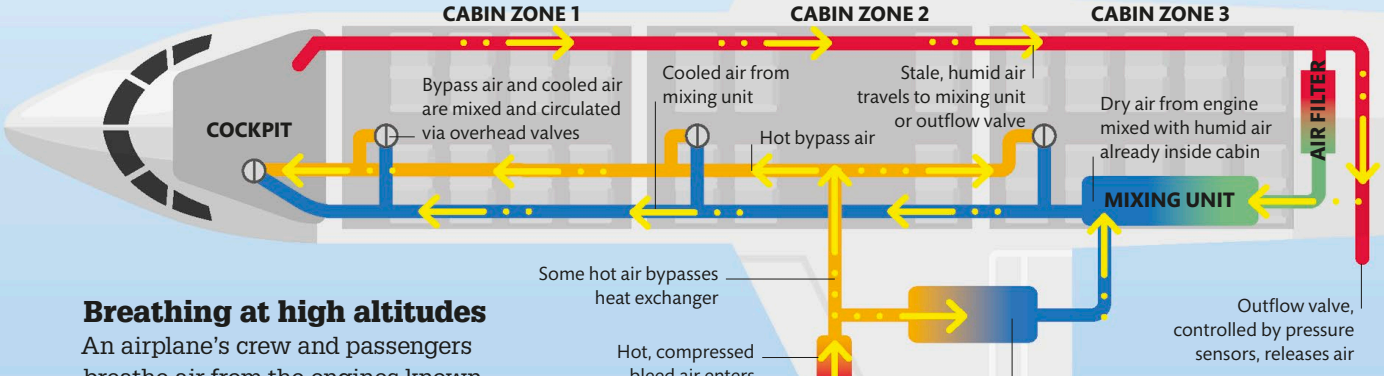
AIR PRESSURE

The air pressure experienced on the ground is caused by the weight of the atmosphere above pressing down. At ground level, the pressure inside and outside an airplane is the same. As it climbs to its cruising altitude, the air pressure outside the airplane falls. The air pressure inside the cabin is maintained at a higher level by a system that pumps air from the engines into the cabin. This ensures that there is enough oxygen for people to breathe.

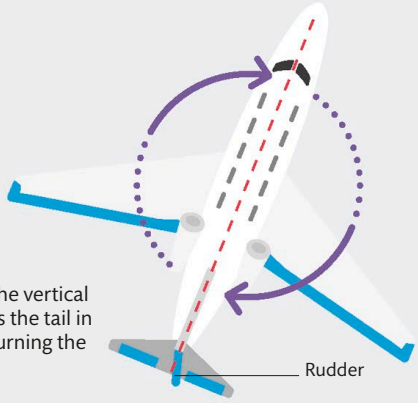




Breathing at high altitudes

An airplane's crew and passengers breathe air from the engines known as bleed air. This superhot air is first chilled by a heat exchanger, called an air cycle machine. It is then mixed with filtered recirculated air and fed into the plane. Stale air escapes through an outflow valve that is opened or closed to regulate the pressure.

Cabin zones
An airliner's cabin is divided into climate zones, each with its own air supply, which can be controlled separately.



Yaw
Swiveling the rudder in the vertical tail fin to one side pushes the tail in the opposite direction, turning the plane's nose left or right.

WHICH IS THE LONGEST SCHEDULED FLIGHT?

A nonstop flight from Singapore to New York covers 9,532 miles (15,341 km) in 17 hours and 25 minutes.